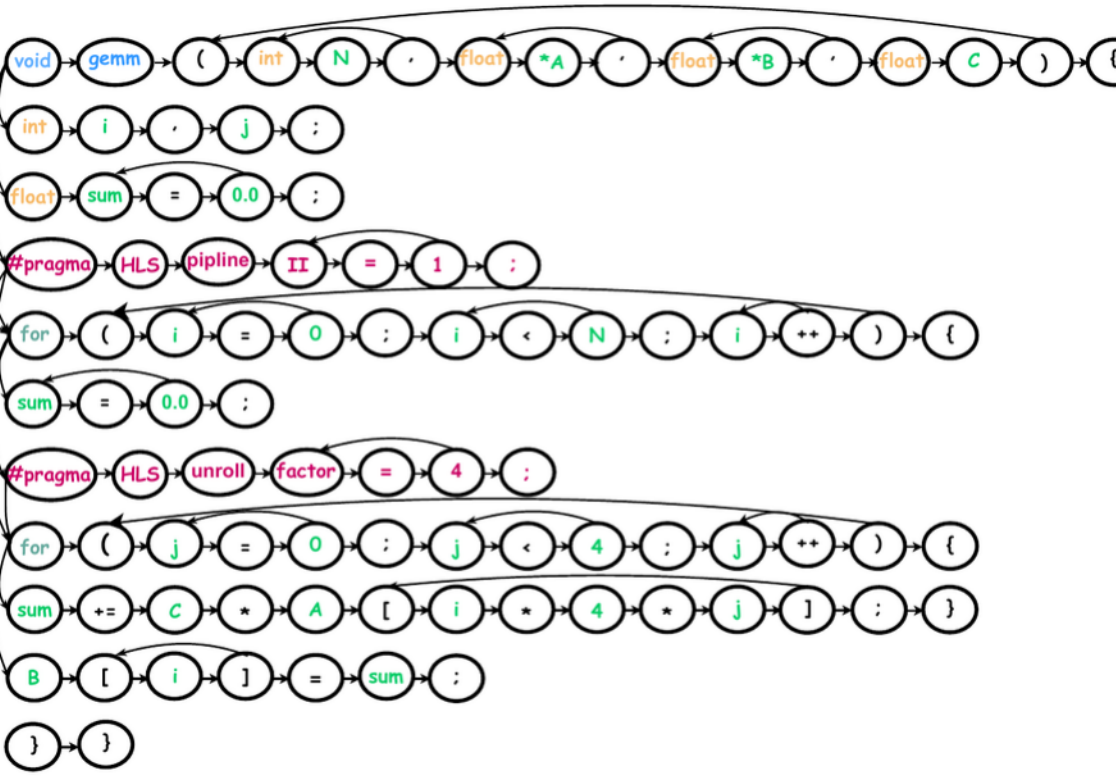

void gemm (int N , float *A , float *B , float C) {
int i , j ;
float sum = 0.0 ;
#pragma HLS pipline II = 1 ;
for (i = 0 ; i < N ; i ++) {
sum = 0.0 ;
#pragma HLS unroll factor = 4 ;
for (j = 0 ; j < 4 ; j ++) {
sum += C * A [i * 4 * j] ; }
B [i] = sum ;
} }